



Essay Questions

Software Construction – Lectures 1 & 2

(Dr. Ehab Essa Exam Style)



Essay Question 1 – Definitions

Define the following terms:

1. **Software Construction**
2. **Static Analysis**
3. **Dynamic Checking**
4. **Immutability**



Model Answer (What gets full marks):

Software Construction:

The process of **building working software** by coding, debugging, testing, and integrating components according to specifications.

Static Analysis:

Analyzing source code **without executing it** to detect errors such as syntax errors, type mismatches, and style violations.

Dynamic Checking:

Checking program behavior **during execution** to detect runtime errors like division by zero and null pointer exceptions.

Immutability:

A property of objects whose state **cannot change after creation**, which improves safety and reasoning about code.

Essay Question 2 – Code & Rules

Consider the following Java code:

```
java Copy code  
  
public class Example {  
    public static void main(String[] args) {  
        String s = "Ahmed";  
        s.concat(" Ali");  
  
        int x = 10 / 0;  
  
        System.out.println(s);  
    }  
}
```

Answer the following questions:

1. Is the code **correct according to software construction principles**?
2. Identify **two problems** in the code.
3. Mention **which concept from Lecture 1 or Lecture 2** is violated in each problem.

Model Answer:

1. **The code is not correct.**
2. **Problems:**
 - The `concat` method does not modify `s`.
 - Division by zero causes a runtime error.
3. **Violated concepts:**
 - `String` is **immutable** (Lecture 2), so `s.concat()` does not change `s`.

- `10 / 0` causes an **ArithmeticException**, detected by **dynamic checking** (Lecture 2).
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Essay Questions

Software Construction – Lecture 3 (Code Review)

(Dr. Ehab Essa Exam Style)



Essay Question 1 – Definitions

Define the following terms:

1. **Code Review**
2. **Pair Programming**
3. **Formal Code Inspection**
4. **Constructive Feedback**



Model Answer (Full Marks):

Code Review:

A systematic examination of source code by developers other than the author to detect defects and improve code quality.

Pair Programming:

A code review technique where two developers work together at one workstation, continuously reviewing each other's code.

Formal Code Inspection:

A structured and planned code review process involving defined roles and documented feedback.

Constructive Feedback:

Clear, respectful comments focused on improving the code, not criticizing the developer.



Essay Question 2 – Code & Rules

Consider the following code snippet:

```
java Copy code  
  
public class Calc {  
    public int calc(int x, int y) {  
        int z = x + y; // calculate result  
        return z;  
    }  
}
```

During a code review, answer the following:

1. Is the code **correct functionally**?
 2. Mention **two improvements** related to **code review best practices**.
 3. Which **code review principles** are applied in your suggestions?
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Model Answer:

1. **Yes, the code is functionally correct.**
2. **Improvements:**
 - Method name `calc` is **not descriptive**.
 - Comment is **unnecessary** because the code is self-explanatory.

3. Applied principles:

- Improve **readability and naming**
 - Remove **redundant comments**
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Essay Questions – Lecture 4: Testing

(Dr. Ehab Essa Exact Exam Style)



Essay Question 1 – Definitions (Testing)

Define the following terms:

1. **Software Testing**
2. **Unit Testing**
3. **Test Case**
4. **Black-box Testing**



Model Answer:

Software Testing:

The process of executing software to detect defects and verify that it behaves as expected.

Unit Testing:

Testing individual units or components of a program in isolation.

Test Case:

A set of inputs, execution conditions, and expected results used to test a program.

Black-box Testing:

Testing based on input and output behavior without knowledge of internal implementation.

Essay Question 2 – Code & Testing Rules

Consider the following method:

```
public static int max(int a, int b) {  
    return a > b ? a : b;  
}
```

Answer the following:

1. Is this method testable?
2. Give **two test cases** for this method.
3. Which **testing principle** is applied when choosing these test cases?

✓ Model Answer:

1. Yes, the method is testable.
 2. Test cases:
 - `max(5, 3) → 5`
 - `max(2, 7) → 7`
 3. **Equivalence partitioning**
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Essay Questions – Lecture 5: Debugging & Exceptions

Essay Question 1 – Definitions (Debugging & Exceptions)

Define the following terms:

1. **Debugging**
2. **Exception**
3. **Checked Exception**
4. **Runtime Exception**

Model Answer:

Debugging:

The process of locating and fixing defects in software.

Exception:

An abnormal event that occurs during program execution and disrupts normal flow.

Checked Exception:

An exception that must be handled or declared at compile time.

Runtime Exception:

An exception that occurs during execution and is not checked at compile time.

Essay Question 2 – Code & Exceptions

Consider the following code:

```
public class Test {  
    public static void main(String[] args) {  
        int[] arr = new int[3];  
        System.out.println(arr[5]);  
    }  
}
```

}

Answer the following:

1. Is this code correct?
2. What exception will be thrown?
3. Is this exception checked or unchecked?
4. Which concept from Lecture 5 applies here?

✓ **Model Answer:**

1. No, the code is incorrect.
 2. **ArrayIndexOutOfBoundsException**
 3. **Unchecked (Runtime Exception)**
 4. **Exception handling / runtime errors**
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Essay Questions

Software Construction – Lecture 6: Specifications

(Dr. Ehab Essa Exact Exam Style)



Essay Question 1 – Definitions

Define the following terms:

1. **Software Specification**
2. **Precondition**
3. **Postcondition**
4. **Behavioral Equivalence**

✓ **Model Answer (Full Marks):**

Software Specification:

A precise description of **what a program or method should do**, without describing how it is implemented.

Precondition:

A condition that must be **true before** a method is called, and it is the **responsibility of the client**.

Postcondition:

A condition that must be **true after** a method finishes execution, and it is the **responsibility of the implementer**.

Behavioral Equivalence:

Two implementations are behaviorally equivalent if a client **cannot distinguish between them** using the specification.

Essay Question 2 – Code & Specifications

Consider the following method specification and implementation:

```
/**
 * Returns the maximum value in the array.
 * @param arr an array of integers
 * @return the largest integer in arr
 */
public static int max(int[] arr) {
    int max = 0;
    for (int x : arr) {
```

```
        if (x > max) {
            max = x;
        }
    }
    return max;
}
```

Answer the following questions:

1. Is the **specification complete and correct**?
2. Does the **implementation satisfy the specification**?
3. Identify **two problems** related to specifications or general rules.
4. Mention the **specification concept** violated in each problem.

✓ **Model Answer (What Dr. Ehab expects):**

1. **No, the specification is incomplete.**
 2. **No, the implementation does not always satisfy the specification.**
 3. **Problems:**
 - The specification does not state a **precondition** (e.g., array not empty).
 - Initializing `max` to `0` fails for arrays with all negative values.
 4. **Violated concepts:**
 - Missing **precondition** in the specification.
 - Implementation does not fulfill the **postcondition**.
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Software Construction – Lecture 7: AI in Software Development

(Dr. Ehab Essa Exact Exam Style)

Essay Question 1 – Definitions

Define the following terms:

1. **Artificial Intelligence (AI)**
2. **Generative AI**
3. **Large Language Model (LLM)**
4. **Prompt Chaining**

Model Answer (Full Marks):

Artificial Intelligence (AI):

The field of computer science that aims to create systems capable of performing tasks that normally require human intelligence.

Generative AI:

A type of AI that can **generate new content** such as text, code, or images based on learned patterns.

Large Language Model (LLM):

A machine learning model trained on large amounts of text to understand and generate human-like language.

Prompt Chaining:

A technique where a complex task is divided into **multiple smaller prompts**, using a **divide-and-conquer** approach.

Essay Question 2 – Prompt & Rules

Consider the following prompt given to an AI tool:

"Write a Java program that reads numbers, sorts them, removes duplicates, handles errors, optimizes performance, writes documentation, and generates test cases."

Answer the following questions:

1. Is this prompt **well-written according to AI software development principles**?
 2. Identify **two problems** with this prompt.
 3. Mention **two improvements** based on Lecture 7 concepts.
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Model Answer (What Dr. Ehab expects):

1. **No, the prompt is not well-written.**
 2. **Problems:**
 - The prompt is **too broad** and overloaded.
 - Multiple tasks are mixed in a single instruction.
 3. **Improvements:**
 - Apply **prompt chaining**.
 - Break the task into **smaller, focused prompts**.
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Essay Questions

Software Construction – Lecture 8: Introduction to Machine Learning

(Dr. Ehab Essa Exact Exam Style)



Essay Question 1 – Definitions

Define the following terms:

1. Machine Learning
 2. Supervised Learning
 3. Unsupervised Learning
 4. Hyperparameter
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Model Answer (Full Marks):

Machine Learning:

A field of computer science that enables systems to **learn from data and improve performance without explicit programming**.

Supervised Learning:

A type of machine learning where the model is trained using **labeled data**.

Unsupervised Learning:

A type of machine learning where the model learns patterns from **unlabeled data**.

Hyperparameter:

A parameter that is **set before training** and controls the learning process of the model.



Essay Question 2 – Code / ML Scenario

Consider the following scenario:

A machine learning model is trained to classify emails as **spam** or **not spam**.
The dataset is split into training, validation, and test sets.

Answer the following questions:

1. Which type of machine learning is used in this problem?
 2. Why should the **test set NOT** be used during training?
 3. What is the role of the **validation set**?
 4. Is choosing the value of k in KNN a model parameter or a hyperparameter?
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✓ **Model Answer (What Dr. Ehab expects):**

1. **Supervised learning**
 2. Because using the test set during training leads to **biased evaluation** and overfitting.
 3. The validation set is used to **tune hyperparameters** and select the best model.
 4. The value of k is a **hyperparameter**.
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Essay Questions

Software Construction – Lecture 9: Introduction to Deep Learning

(Dr. Ehab Essa Exact Exam Style)

Essay Question 1 – Definitions

Define the following terms:

1. **Deep Learning**
2. **Artificial Neural Network (ANN)**
3. **Activation Function**
4. **Weight**

Model Answer (Full Marks):

Deep Learning:

A subfield of machine learning that uses **multi-layer neural networks** to automatically learn feature representations from data.

Artificial Neural Network (ANN):

A computational model inspired by the human brain, composed of **interconnected neurons organized in layers**.

Activation Function:

A function applied to a neuron's output to introduce **non-linearity** into the network.

Weight:

A numerical value that represents the **strength of the connection** between neurons.

Essay Question 2 – Architecture / Conceptual Analysis

Consider the following neural network description:

A neural network consists of an input layer, one hidden layer, and an output layer.

Each neuron computes a weighted sum of its inputs, adds a bias, then applies an activation function.

Answer the following questions:

1. What type of model is described above?
 2. What is the role of the **hidden layer**?
 3. Why are **activation functions** necessary?
 4. Why can a **single-layer perceptron** solve only certain types of problems?
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✓ **Model Answer (What Dr. Ehab expects):**

1. **Artificial Neural Network (Multi-layer network)**
 2. The hidden layer learns **intermediate feature representations**.
 3. Activation functions introduce **non-linearity**, allowing the network to model complex relationships.
 4. A single-layer perceptron can solve only **linearly separable problems**.
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